```
sns.set_theme(style="whitegrid")

print("All libraries imported successfully!")
```

```
All libraries imported successfully!
```

```
display(df.describe(include="all"))
```

	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner	mileage(km/ltr/kg)	engine	max_power	seats
4	Maruti Swift VXI BSIII	2007	130000	120000	Petrol	Individual	Manual	First Owner	16.10	1298.0	88.2	5.0

	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner	mileage(km/ltr/kg)	engine	max_power	seats
8123	Hyundai i20 Magna	2013	320000	110000	Petrol	Individual	Manual	First Owner	18.50	1197.0	82.85	5.0
8124	Hyundai Verna CRDi SX	2007	135000	119000	Diesel	Individual	Manual	Fourth & Above Owner	16.80	1493.0	110	5.0
8125	Maruti Swift Dzire ZDi	2009	382000	120000	Diesel	Individual	Manual	First Owner	19.30	1248.0	73.9	5.0
8126	Tata Indigo CR4	2013	290000	25000	Diesel	Individual	Manual	First Owner	23.57	1396.0	70	5.0
8127	Tata Indigo CR4	2013	290000	25000	Diesel	Individual	Manual	First Owner	23.57	1396.0	70	5.0

```
Dataset shape: (8128, 12)
Column names:
['name', 'year', 'selling_price', 'km_driven', 'fuel', 'seller_type', 'transmission', 'owner', 'mileage(km/ltr/kg)', 'engine', 'max_power', 'seats']
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8128 entries, 0 to 8127
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   name                8128 non-null   object
 1   year                8128 non-null   int64
 2   selling_price       8128 non-null   int64
 3   km_driven           8128 non-null   int64
 4   fuel                8128 non-null   object
 5   seller_type         8128 non-null   object
 6   transmission        8128 non-null   object
 7   owner               8128 non-null   object
 8   mileage(km/ltr/kg)  7907 non-null   float64
 9   engine              7907 non-null   float64
 10  max_power           7913 non-null   object
 11  seats               7907 non-null   float64
dtypes: float64(3), int64(3), object(6)
memory usage: 762.1+ KB
```

	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner	mileage(km/ltr/kg)	engine	max_power	seats
count	8128	8128.000000	8.128000e+03	8.128000e+03	8128	8128	8128	8128	7907.000000	7907.000000	7913	7907.000000
unique	2058	NaN	NaN	NaN	4	3	2	5	NaN	NaN	320	NaN
top	Maruti Swift Dzire VDI	NaN	NaN	NaN	Diesel	Individual	Manual	First Owner	NaN	NaN	74	NaN

```
Missing values:
name                      0
year                      0
selling_price             0
km_driven                 0
fuel                      0
seller_type               0
transmission              0
owner                     0
mileage(km/ltr/kg)      221
engine                  221
max_power               215
seats                   221
dtype: int64

Duplicate rows: 1202

Dataset shape after removing duplicates: (6926, 12)
```

```
Updated columns:
['name', 'year', 'selling_price', 'km_driven', 'fuel', 'seller_type', 'transmission', 'owner', 'mileage_km_ltr_kg', 'engine', 'max_power', 'seats']
```

```
        df[col] = df[col].str.lower()

print("Categorical values standardized.")
```

```
Categorical values standardized.
```

```
print( Numeric columns converted. )
```

```
Numeric columns converted.
```

```
print("Dataset shape after target cleaning:", df.
```

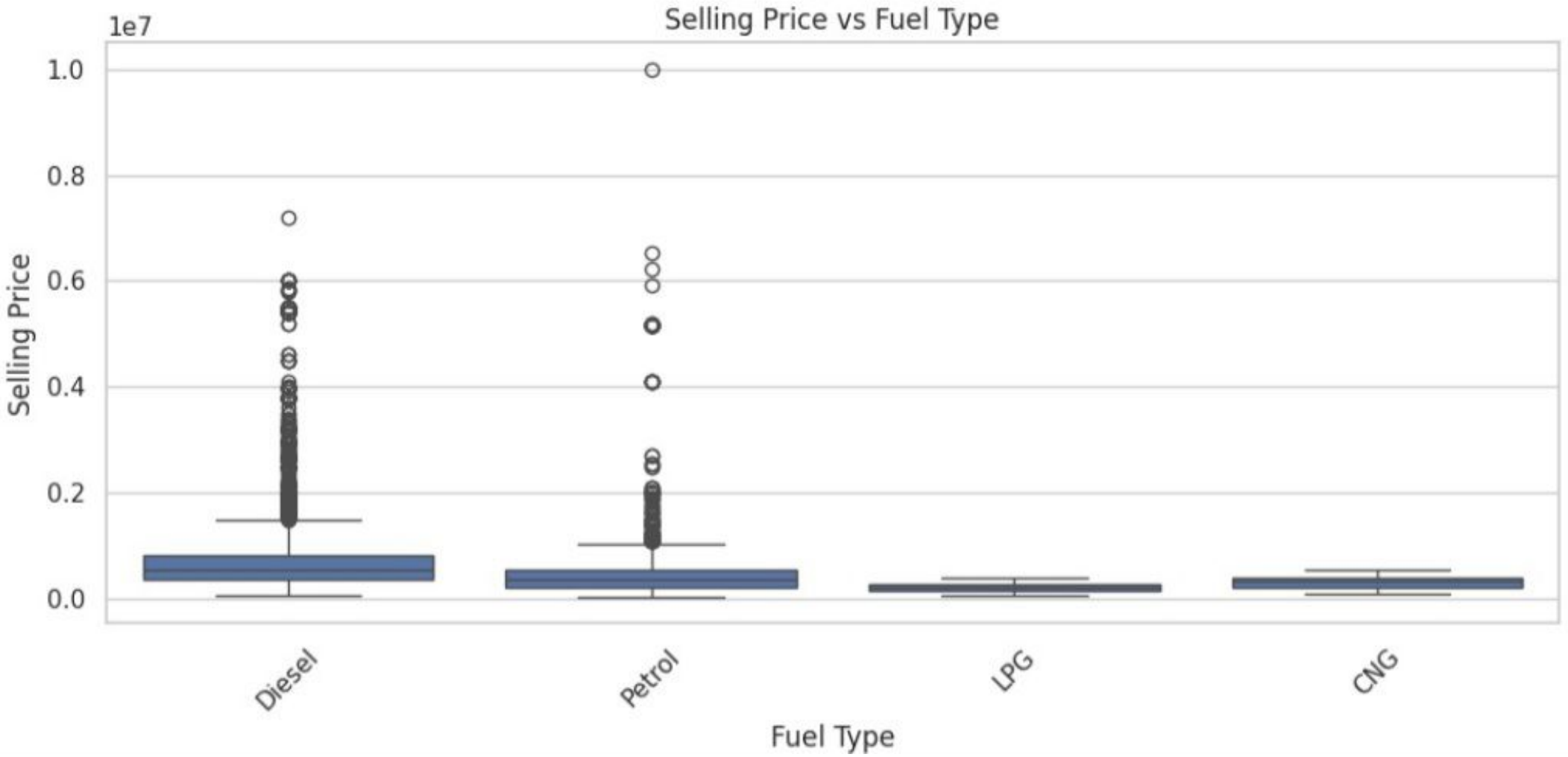
```
Dataset shape after target cleaning: (6926, 12)
```

```
            if 'year' in df.columns
```

```
...        year  car_age
    0  2014.0     12.0
    1  2014.0     12.0
    2  2006.0     20.0
    3  2010.0     16.0
    4  2007.0     19.0
```

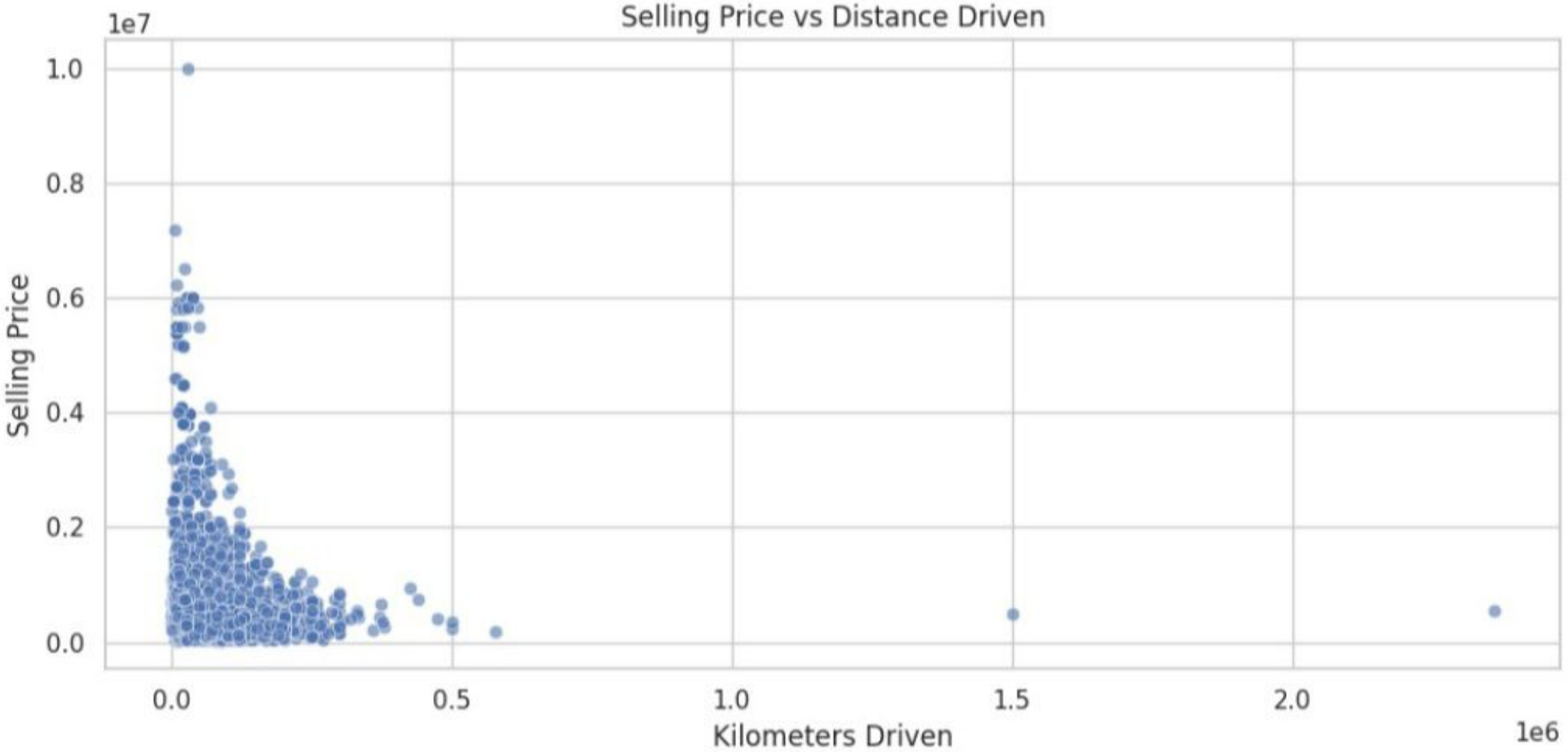
```
Car brands extracted successfully.
                                name    brand
0         Maruti Swift Dzire VDI    maruti
1  Skoda Rapid 1.5 TDI Ambition     skoda
2       Honda City 2017-2020 EXi     honda
3     Hyundai i20 Sportz Diesel   hyundai
4         Maruti Swift VXI BSIII    maruti
```

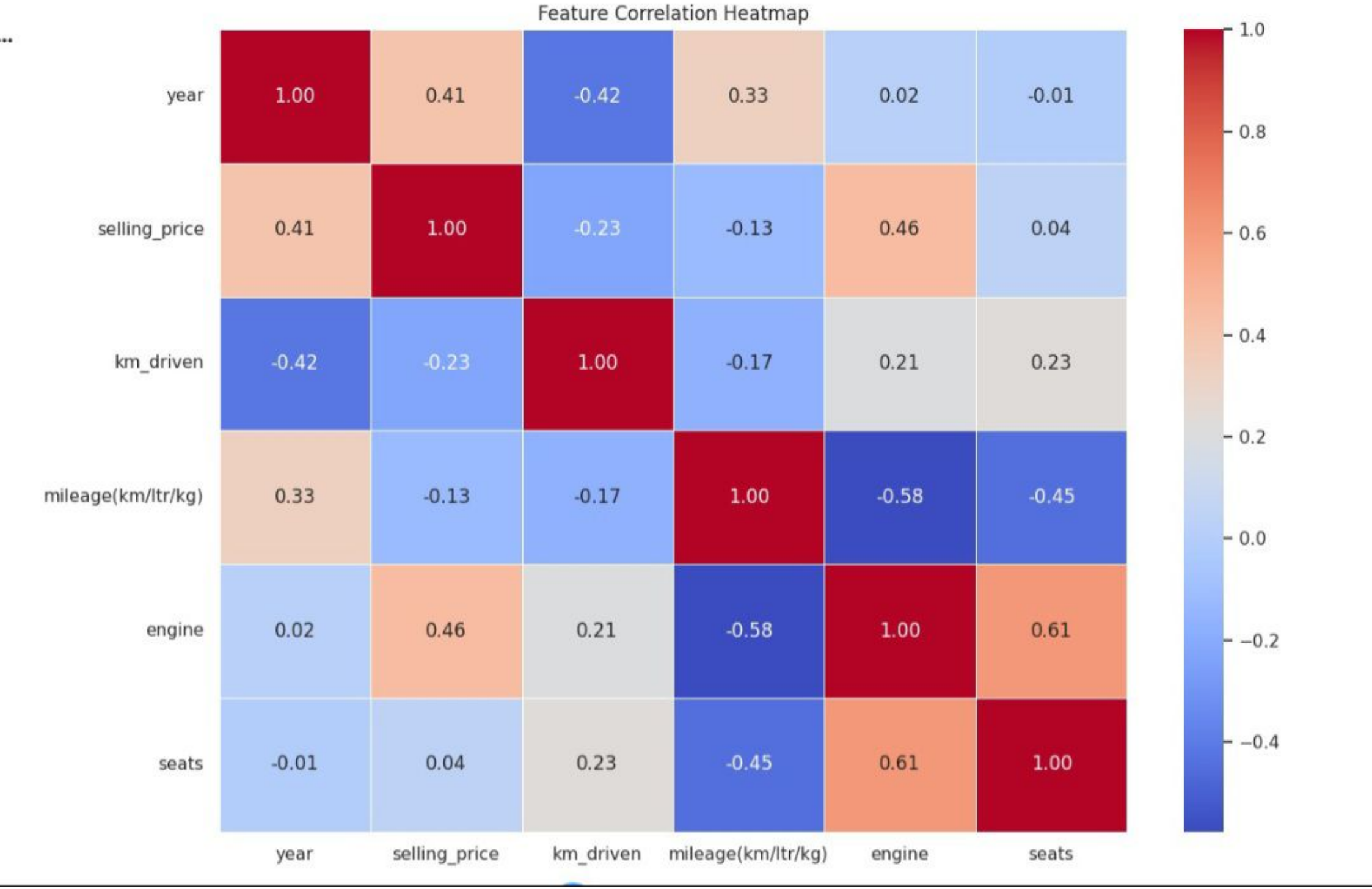

Distribution of Used Car Selling Prices
Frequency
3500
3000
2500
2000
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
1e7
Selling Price


Selling Price vs Fuel Type
1e7
1.0
0.8
0.6
0.4
0.2
0.0
Selling Price
Diesel
Petrol
LPG
CNG
Fuel Type

```
else:
    print("Car age feature unavailable
```

```
Car age feature unavailable.
```


Selling Price vs Distance Driven
1e7
1.0
0.8
0.6
0.4
0.2
0.0
Selling Price
0.0
0.5
1.0
1.5
2.0
1e6
Kilometers Driven


Feature Correlation Heatmap
year
selling_price
km_driven
mileage(km/ltr/kg)
engine
seats
1.00 0.41 -0.42 0.33 0.02 -0.01
0.41 1.00 -0.23 -0.13 0.46 0.04
-0.42 -0.23 1.00 -0.17 0.21 0.23
0.33 -0.13 -0.17 1.00 -0.58 -0.45
0.02 0.46 0.21 -0.58 1.00 0.61
-0.01 0.04 0.23 -0.45 0.61 1.00
year
selling_price
km_driven
mileage(km/ltr/kg)
engine
seats
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

```
display(X.head())
```

```
Features used:
['km_driven', 'engine', 'max_power', 'seats', 'fuel', 'transmission', 'seller_type', 'owner']

Feature shape: (8128, 8)
Target shape: (8128,)
```

	km_driven	engine	max_power	seats	fuel	transmission	seller_type	owner
0	145500	1248.0	74	5.0	Diesel	Manual	Individual	First Owner
1	120000	1498.0	103.52	5.0	Diesel	Manual	Individual	Second Owner
2	140000	1497.0	78	5.0	Petrol	Manual	Individual	Third Owner
3	127000	1396.0	90	5.0	Diesel	Manual	Individual	First Owner
4	120000	1298.0	88.2	5.0	Petrol	Manual	Individual	First Owner

```
Training samples: 6502
Testing samples: 1626
```

```
)

print("Preprocessing pipeline created successfully.")
```

```
Numerical features: ['km_driven', 'engine', 'seats']
Categorical features: ['max_power', 'fuel', 'transmission', 'seller_type', 'owner']
Preprocessing pipeline created successfully.
```

```
print("Linear Regression model trained successfully!")
```

```
Linear Regression model trained successfully!
```

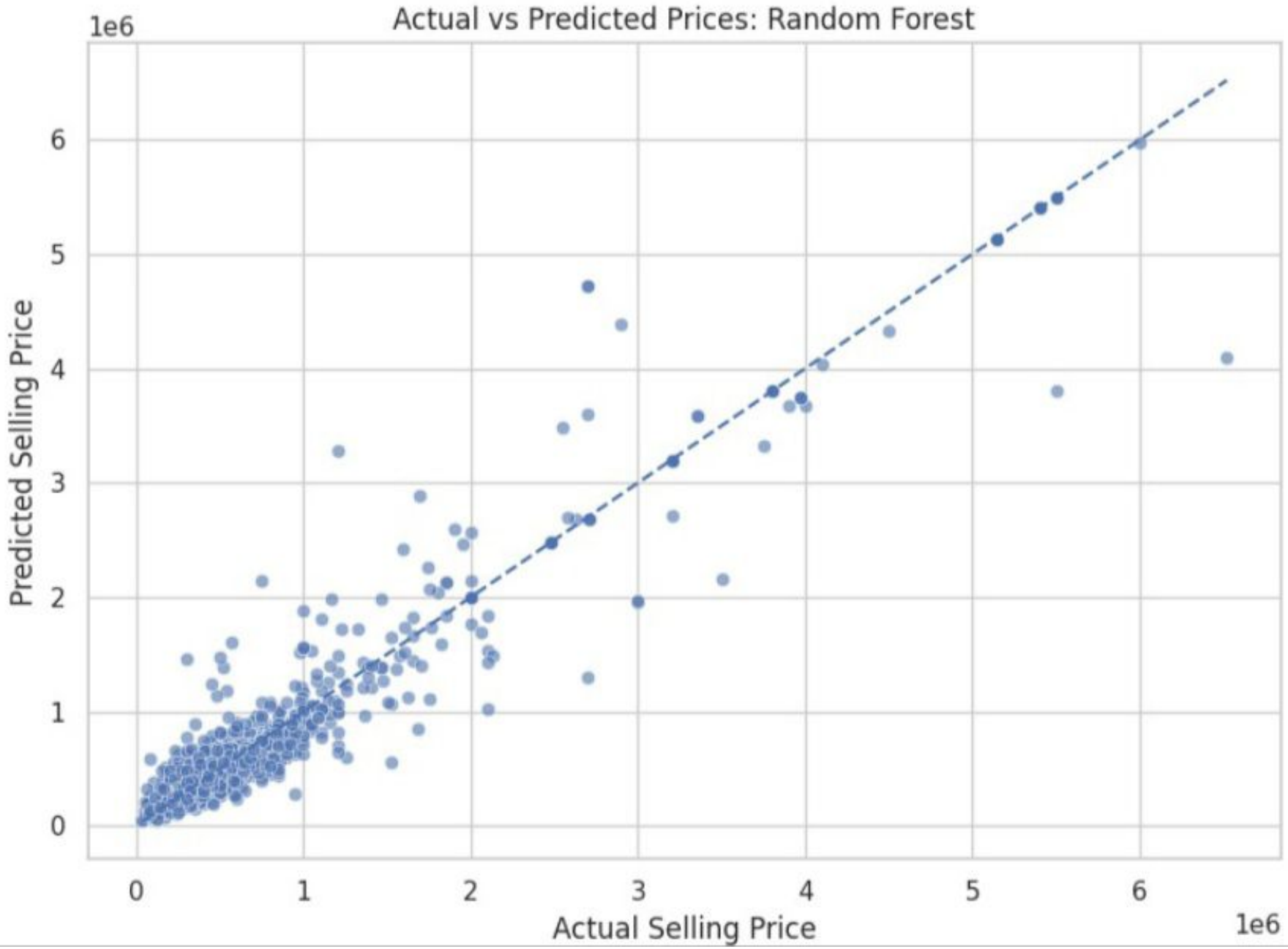
```
Random Forest model trained successfully!
```

	Model	MAE	RMSE	R2 Score
0	Linear Regression	302166.424	506131.628	0.609
1	Random Forest	106273.876	212729.204	0.931

```
Best model based on test RMSE: Random Forest

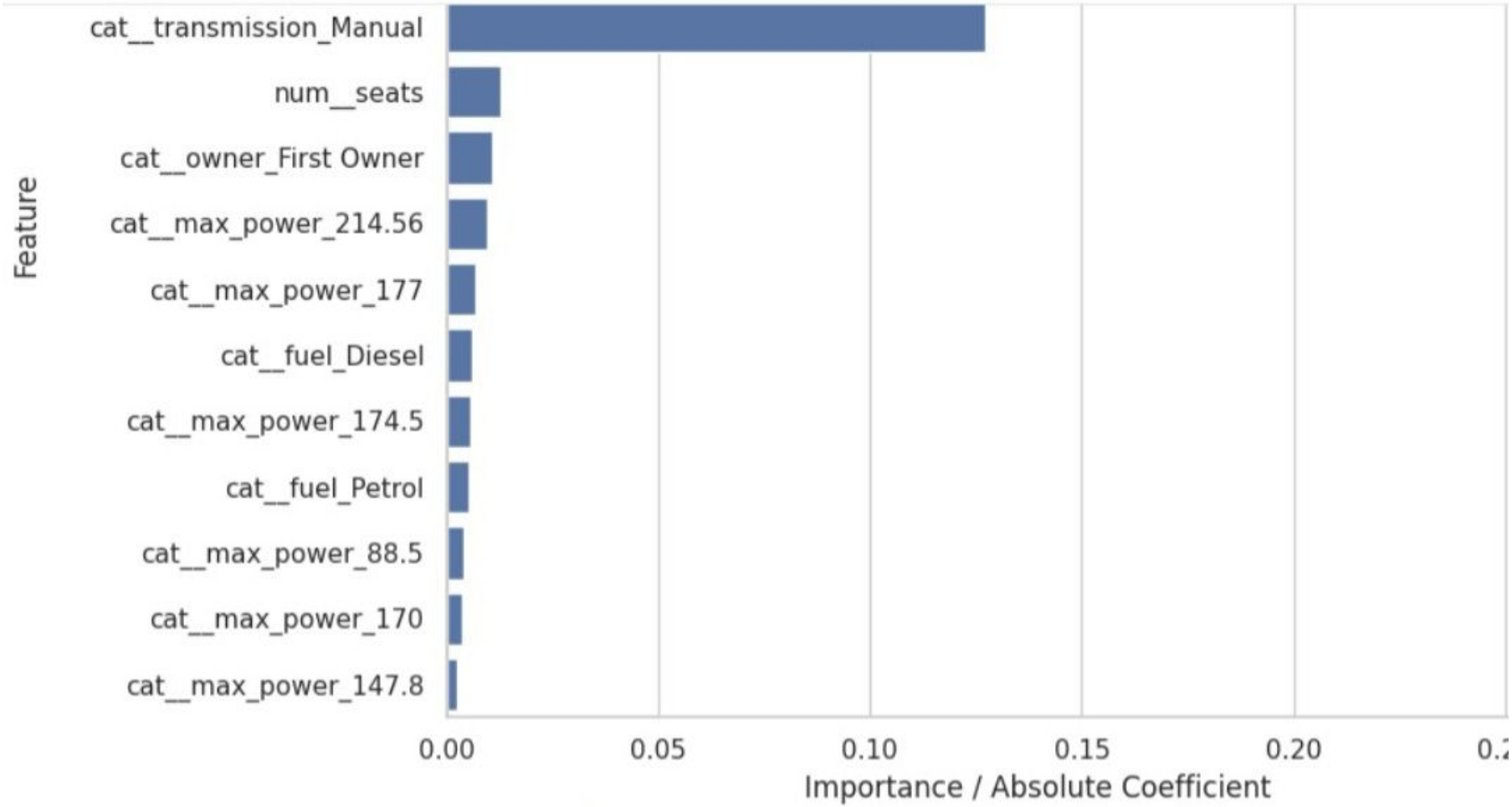
Best model evaluation:
           Model        MAE        RMSE  R2 Score
Random Forest 106273.876 212729.204     0.931
```

```
plt.show()
```


Actual vs Predicted Prices: Random Forest
1e6
Predicted Selling Price
0
1
2
3
4
5
6
Actual Selling Price
1e6

```
plt.tight_layout()
plt.show()

display(importance_df)
```



	Feature	Importance
1	num__engine	0.265660
135	cat__max_power_190	0.205756
0	num__km_driven	0.155699
315	cat__transmission_Automatic	0.135509
316	cat__transmission_Manual	0.127170
2	num__seats	0.012698

Car details:

	km_driven	engine	max_power	seats	fuel	transmission	seller_type	owner
1971	110000	1493.0	100	5.0	Petrol	Manual	Individual	Third Owner

Actual selling price: ₹198,000.00
Predicted selling price: ₹168,453.66
Absolute prediction error: ₹29,546.34

```
Model saved successfully!
```